T-M-580
July 28, 1965

SUPERSEDING
Fed. Spec. T-S-637a
January 19, 1955

FEDERAL SPECIFICATION

MOP, WET, COTTON, (WITH FIXED HANDLE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cotton mops with fixed handles for swabbing floors and decks.

1.2 Classification. Cotton cleaning mops covered by this specification shall be of one type and size (see fig. 1).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

T-R-675—Rope, Yarn, and Twine; Hemp.
T-T-881—Twine; Cotton, Seine.
UU-P-268—Paper, Kraft, Untreated, Wrapping.
CCC-C-429—Cloth, Osnaburg, Cotton.
CCC-C-467—Cloth, Jute (or Kenaf), Burlap.
CCC-T-191—Textile Test Methods.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-W-6110—Wood, Determination of Moisture Content of.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall conform to the applicable specifications and requirements specified herein. Where materials are not definitely specified, they shall be of

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the best quality used for the purpose in commercial practice.

3.1.1 *Wood.* Wood for handles shall be one of the following species: ash, beech, birch, cherry, elm, hickory, maple, pecan, ramin, red or white oak, Douglas-fir, or Southern yellow pine. The wood shall be free of knots, shakes, wane, worm holes, heart center, rot, dote, and brash, and shall have a moisture content of not less than 8 percent nor more than 12 percent when tested in accordance with 4.2.1.

3.1.2 *Swab yarn.* The swab yarn shall be not less than 8-ply bleached or unbleached cotton yarn measuring not less than 600 yards to the pound and having a breaking strength of not less than 15 pounds when tested in accordance with methods 4010 and 4102, respectively, of CCC-T-191.

3.1.3 *Wrapping twine.* Wrapping twine shall be a cotton seine twine conforming to type I of T-T-881, except that the twine shall be commercial No. 60, 3-ply, 20 strands per ply, having a minimum breaking strength of 108 pounds and measuring approximately 100 yards of twine per pound. As an alternate to the cotton seine twine, medium tarred marline conforming to type IV, class 1, of T-R-675 may be used.

3.2 **Design.**

3.2.1 *Handles.* Handles shall be made of a wood species in 3.1.1 and shall not have a cross grain or spiral grain sloping more than 1 inch in 15 inches. Handles shall be straight, rounded at both ends, and of uniform diameter. Handles shall be 48 ± 1/2 inch in length and not less than 15/16 inch nor more than 1-5/16 inches in diameter, except that handles made of Douglas-fir or Southern yellow pine shall be not less than 1 inch in diameter.

3.2.1.1 One and one fourth inches from the swab end, the handle shall be notched down to a diameter of 1/8 to 3/16 inch less than its nominal diameter and then, for a length of approximately 2 inches, shall be tapered up to the full diameter of the handle. A 7/32-inch diameter hole shall be drilled through the center of and perpendicular to the axis of the handle approximately 2-1/4 inches from the swab end.

3.2.2 *Swab.* The swab shall consist of not less than 1-1/2 pounds of cotton yarn specified in 3.1.2, cut in strands not less than 38 inches in length. The finished swab end shall measure not less than 17 inches in length and the ends of the strands shall be evenly trimmed.

3.3 **Construction.** The swab shall be firmly attached to the handle with twine specified in 3.1.3.

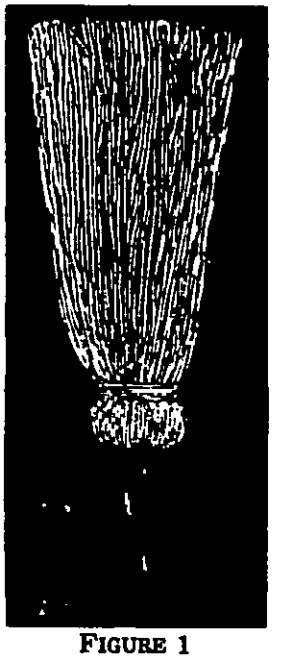


FIGURE 1

3.3.1 The twine shall be secured to the handle by passing the end through the hole in the handle and securing the end by laying four turns of twine over it. The handle shall then be centered over the cotton strands and the strands uniformly distributed around the handle periphery. The assembly shall be secured with six turns of the wrapping twine; two turns shall be made at the top of the tapered notch, two at an angle, and two at the bottom of the notch. The ends of the strands toward the handle

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shall be doubled over the six turns of twine and the strand ends pulled even. Eight additional turns of twine shall then be laid around the outside of the swab and the end of the twine secured by drawing it under the eight turns.

3.4 Finish. Each handle shall be sanded smooth throughout its length and completely coated with a clear lacquer, varnish or other clear organic finish. Two applications of the coating material shall be applied to handles made of Douglas-fir, Southern yellow pine, elm, and red and white oak. One application of coating material is acceptable for handles made of ash, beech, birch, hickory, maple, pecan, and ramin; however, the coating shall be of a thickness to impart a pronounced gloss to the surface of the handle.

3.5 Workmanship. Wood handles shall be free of defects in 3.1.1 and shall be free of raised grain or other defects that may cause personal injury or affect serviceability. The mop strands shall be evenly trimmed and tightly bound.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with this specification or referenced subsidiary specifications.

4.2.1 *Moisture content of wood.* The moisture content of wood (see 3.1.1) shall be determined in accordance with MIL-W-6110 after handle fabrication but prior to fabrication of the mop. Ten sample handles to be used in the end-item inspection lot shall be tested. Failure of any sample to meet the moisture content requirement shall be cause for rejection of the lot of handles.

4.2.2 *Swab yarn.* Each lot of swab yarn for use on the contract or order shall be tested for compliance with 3.1.2. The sample unit shall be one pound of yarn. Ten samples shall be selected at random from the lot and tested in accordance with CCC-T-191, method 4010 for the yards per pound and method 4102 for the tensile strength. The average of the ten results for the sample size shall meet the requirements of 3.1.2.

4.3 Sampling and inspection for lot acceptance. Sampling and inspection shall be performed in accordance with MIL-STD-105. For sampling purposes, a lot shall consist of all mops submitted for delivery at one time.

4.3.1 *Inspection of the end item.* Each lot shall be examined in accordance with 4.3.1.1 and 4.3.1.2 and tested in accordance with 4.3.1.3. The sample unit shall be 1 mop and the lot size shall be expressed in units of mops.

4.3.1.1 *Visual examination.* Each sample unit shall be examined for visual defects in table I. Inspection level II shall apply and the acceptable quality level (AQL) shall be 4.0 defects per hundred units.

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TABLE I—*Visual defects in the end item*

Examine	Defect
Materials:	Not wood species, yarn, or twine specified.
Handle:	Not free of knots, shakes, wane, worm holes, heart center, rot, dote or brash.
	Cross grain or spiral grain sloping more than one inch in 15 inches.
	Not round at both ends.
	Not straight.
	Diameter not uniform.
	Swab end not notched and tapered.
	Twine hole missing.
	Not coated with one or two coats, as applicable, of clear varnish, lacquer or clear organic finish.
	Not a glossy appearance (as applicable).
	Area void of coating.
	Not smooth throughout its length.
Swab:	Layover not equal length.
	Twine not tightly wound.
	Less than eight turns of wrapping twine on outer tie.
	End of twine not secured by drawing under outer turns.
	Deck yarn less than 8-ply yarn.
	Yarn not uniformly distributed around periphery of handle.
Workmanship:	Handle not free of raised grain.
	Strands not evenly trimmed.
	Any defect that may cause personal injury.

4.3.1.2 *Dimensional examination.* Each sample unit shall be examined for dimensional defects in table II. Inspection level S-2 shall apply and the AQL shall be 2.5 defects per hundred units.

TABLE II—*Dimensional defects in the end item*

Examine	Defect
Handle:	
Length	Less than 47-1/2 inches or more than 48-1/2 inches.
Diameter	Less than 15/16 inch or 1 inch, as applicable, or more than 1-5/16 inches.
Notch	Depth of notch less than 1/8 inch or more than 3/16 inch.
Swab:	
Length	Less than 17 inches.

4.3.1.3 *Testing of the end item.* Each sample unit shall be tested in accordance with 4.4 to determine compliance with yarn weight requirements in 3.2.2. Inspection level S-2 shall apply and the AQL shall be 2.5 defects per hundred units.

4.3.2 *Examination of the preparation for delivery.* The packaging, packing, and marking shall be examined to determine compliance with section 5 and the special requirements in the contract or order. The lot size shall be expressed in units of shipping bundles and the sample unit shall be 1 shipping bundle fully prepared for delivery. Inspection level S-2 shall apply and lot acceptance determined at an AQL of 2.5 defects per hundred units. Examination shall be made for the defects in table III.

TABLE III—*Defects in the preparation for delivery*

Examine	Defect
Marking (packages and shipping bundles)	Not in accordance with specification requirements.
	Omitted, incomplete, incorrect or illegible.
	Improper size, location, sequence, or method of application.
Packaging	Less than 10 mops per bundle.
	Mops not arranged with 5 heads at each end of bundle.
	Mopheads not tied with marline twine.
	Less than 4 turns of marline at each end.
Packing	Not wrapped with kraft paper.
	Not overwrapped with burlap or osnaburg, as applicable.

4.4 **Testing of swab yarn.** Samples selected in accordance with 4.3.1.3 shall have the swab yarn removed. The weight of yarn in the swab shall then be determined to the nearest ounce.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 should be referred to for definitions and application of the various levels of packaging and packing.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Mops shall be bundled in

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units of 10, with 5 swab heads at each end of the bundle. The swab heads shall be tightly bound with not less than four turns of marline.

5.1.2 *Level C.* Mops shall be packaged in accordance with the manufacturer's standard commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Mops, packaged as specified in 5.1, shall have the swab ends wrapped in not less than 60 pound kraft paper conforming to UU-P-268. The wrap shall be secured in place with either twine or gummed tape. Four wrapped packages shall be overwrapped with osnaburg or burlap cloth conforming to CCC-C-429 or CCC-C-467 respectively, and securely tied with twine.

5.2.2 *Level B.* Mops, packaged as specified in 5.1, shall have the mop heads wrapped with not less than 60 pound kraft paper conforming to UU-P-268 and secured with 3-inch wide gummed kraft tape. Two bundles shall then be securely tied with twine.

5.2.3 *Level C.* Mops, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, interior wrapped bundles and the outer wrapping of each shipping bundle shall be marked in accordance with MIL-STD-129 for military procurement and in accordance with Fed. Std. No. 123 for civil agency procurement.

6. NOTES

6.1 Intended use. Wet mops are intended for the wet-cleaning of decks of ships and other flooring.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of the applicable level of packaging and packing (see 5.1 and 5.2).

6.3 Transportation description. The transportation description and minimum weights applicable to this commodity are:

Rail:

Mops, not otherwise indexed by name.
Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Mops, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.4 Supersession. Implementation of Federal Item Identification Guides for Supply Cataloging necessitated a change in the item name from "Swab, Deck" to "Mop, Wet, Cotton (with Fixed Handle)", and a change in document number to T-M-580.

CUSTODIANS:

Army—GL
Navy—SA
Air Force—69

Review activities:

Army—GL, MO
Navy—SA
Air Force—69

User activities:

Navy—CG, YD, MC
Air Force—26

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Preparing Activity:

Navy—SA

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